0ICPC455

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Machine Learning Lab

Quality Laboratory Manual

Experiment No. 1

Study and Implementation of Simple Linear Regression

Course Instructor -

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Experiment No. 1

Title of Experiment: To study and implement Simple Linear Regression

Aim of Experiment: To implement and understand the working of Simple Linear Regression, basic predictive modeling technique in Machine Learning

System Requirements – Win 8 and above OS, 4GB RAM, 2.33 GHz Processor

Software/s Needed for Experiment – Jupyter Notebook/ Anaconda Navigator/ Google Colaboratory/ Spyder, Python 3.x [With libraries such as Numpy, Pandas, Matplotlib and Scikit-Learn]

Experiment Outcomes –

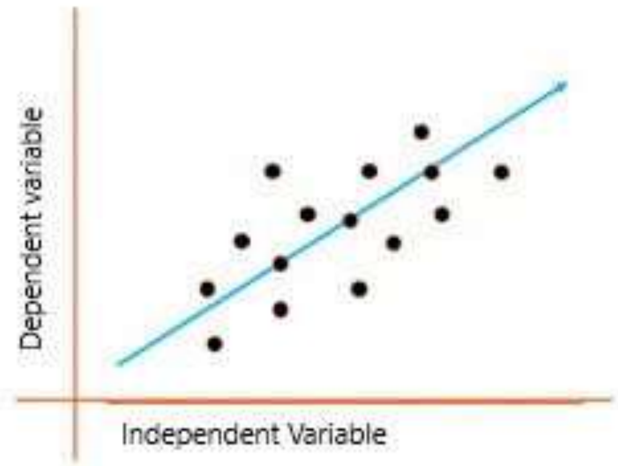
1. Understand the principles and applications of simple linear regression
2. Differentiate between simple and multiple linear regression
3. Analyze and implement the difference between univariate and multivariate analysis
4. To implement linear regression in Python

Theory –

Simple Linear Regression is a statistical method that models the relationship between two variables by fitting a linear equation to observed data. One variable is considered the dependent variable (or **response**), and the other is the independent variable (or predictor).

Linear regression predicts the relationship between two variables by assuming they have a straight-line connection. It finds the best line that minimizes the differences between predicted and actual values. Used in fields like economics and finance, it helps analyze and forecast data trends. Linear regression can also involve several variables (multiple linear regression) or be adapted for yes/no questions (logistic regression).

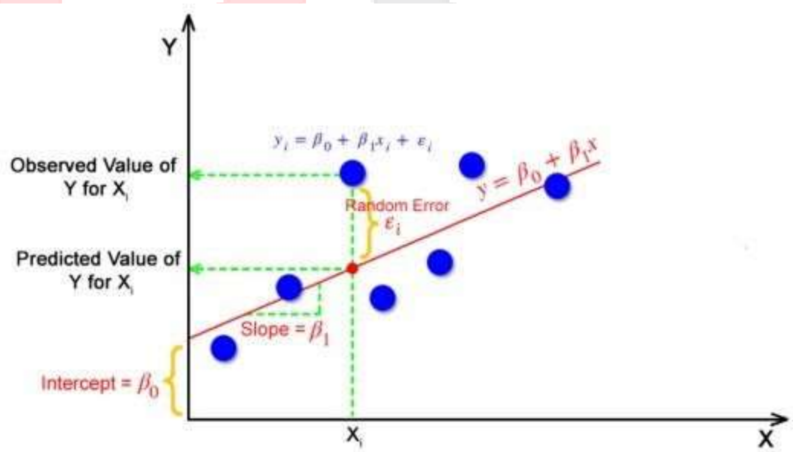
Linear regression is a quiet and the simplest statistical regression technique used for predictive analysis in machine learning. Linear regression shows the linear relationship between the independent(predictor) variable i.e. X-axis and the dependent (output) variable i.e. Y-axis, called linear regression. If there is a single input variable X (independent variable), such linear regression is simple linear regression.



The equation of a straight line is given by:
$y = \beta_0 + \beta_1 \times x$
where:

- y is the dependent variable.
- x is the independent variable.
- β_0 is the intercept of the line.
- β_1 is the slope of the line

The goal of linear regression is to find the values of β0 and β1 that minimize the sum of squared errors (SSE) between the observed values and the values predicted by the model



Procedure to implement Simple Linear Regression –

1. **Data Collection** – Collect or generate a dataset with tow numerical variables. For this experiment use a simple dataset representing the relationship between the number of study hours and exam scores, CO2 emission of a car, salary of an employee based on years of experience, etc…

2. **Data Processing** – Load the dataset into Pandas DataFrame. Perform any necessary preprocessing such as handling missing values or outliers

3. **Plotting the Data** – Plot the data points on a 2D graph to visualize the relationship between the variables. Use Matplotlib to create the scatter plot

4. **Implementing Simple Linear Regression** – Use scikit-learn library to fit a simple linear regression model. Split the dataset into training and test sets. Train the model using the training data

5. **Model Prediction** – Use the trained model to make predictions on the test data. Plot the regression line along with the data points to visualize the fit

6. **Model Evaluation** – Calculate performance metrics such as Mean Squared Error (MSE) and the coefficients of determination (R^2) to evaluate the models accuracy

Sample Code –

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Step 1: Data Collection (Generating a simple dataset)
# Example dataset: Study Hours vs. Exam Scores
data = {
    'Hours': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Scores': [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
}

df = pd.DataFrame(data)

# Step 2: Data Preprocessing
# (No missing values in this simple dataset)

# Step 3: Plotting the Data
plt.scatter(df['Hours'], df['Scores'], color='blue')
plt.xlabel('Hours Studied')
plt.ylabel('Scores Achieved')
plt.title('Study Hours vs. Exam Scores')
plt.show()

# Step 4: Implementing Simple Linear Regression
X = df[['Hours']]  # Independent variable
```

```
y = df['Scores']  # Dependent variable

# Splitting the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)

# Fitting the Simple Linear Regression model
regressor = LinearRegression()
regressor.fit(X_train, y_train)

# Step 5: Model Prediction
y_pred = regressor.predict(X_test)

# Plotting the regression line
plt.scatter(X_test, y_test, color='blue')
plt.plot(X_test, y_pred, color='red')
plt.xlabel('Hours Studied')
plt.ylabel('Scores Achieved')
plt.title('Regression Line')
plt.show()

# Step 6: Model Evaluation
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print(f"Mean Squared Error: {mse}")
print(f"R^2 Score: {r2}")

# Step 7: Conclusion
print(f"Intercept (β0): {regressor.intercept_}")
print(f"Slope (β1): {regressor.coef_[0]}")
```

Observations –

- Record the predicted scores for the test data
- Observe the regression line's fit to the data point

Conclusion –

Hence, the experiment provides students with hands-on experience in implementing and analyzing a Simple Linear Regression model, a foundational technique in data science and Machine Learning

References –

a. **Textbook –**
    i. Machine Learning with Python – An approach to Applied ML – Abhishek Vijayvargiya, BPB Publications, 1st Edition 2018
    ii. Machine Learning, Tom Mitchell, McGraw Hill Education, 1st Edition 1997

b. **Online references –**
    i. https://www.analyticsvidhya.com/blog/2021/10/everything-you-need-to-know-about-linear-regression/
    ii. https://www.javatpoint.com/simple-linear-regression-in-machine-learning
    iii. https://www.24tutorials.com/machine-learning/simple-linear-regression/

Expected Oral Questions –

1. What is Simple Linear Regression?
2. What are the key components of a Simple Linear Regression?
3. Explain the difference between the dependent variable and the independent variable
4. What is the purpose of using a regression line in Simple Linear Regression?
5. What assumptions are made in Simple Linear Regression?
6. What are the parameters of a Linear Regression?
7. What is the formula for a Linear Regression line?
8. What are the applications for Linear Regression?
9. What is Linear Regression algorithm?
10. State a basic example for Linear Regression

FAQ's in Interview –

1. What distinguishes simple linear regression from multiple linear regression?
2. Is it necessary to visualize the data when you have fitted a line? Justify your any.
3. How do you interpret a linear regression model?
4. Is linear regression suitable for time series analysis?
5. Cal linear regression be used for representing quadratic equations?